GUIDE DE TESTS

SmartLogix Prioritizer with IA

Scenarios complets - Tous utilisateurs - Toutes fonctionnalites

Version : 1.0

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Application : SmartLogix / SmartTask

Frontend : http://localhost:4200

Backend API : http://localhost:9090

Service IA : http://localhost:8000

Swagger : http://localhost:9090/swagger-ui.html

1. INTRODUCTION ET PREREQUIS

1.1 Objectif du guide

Ce guide permet de tester integralement SmartLogix : authentification, 5 roles, workflows commande-livraison, intelligence artificielle, automation, chatbot, incidents et securite. Chaque scenario indique les etapes UI, le resultat attendu et les cas d'echec.

1.2 Demarrage des services

Terminal 1 - PostgreSQL

PostgreSQL actif sur port 5432, base smarttask_db creee.

Terminal 2 - Backend Spring Boot

cd "Backend Smarttask/SmartTask" puis .\gradlew bootRun

Attendre : SmartTask demarre sur port 9090, Flyway migrations OK, DataInitializer seed.

Terminal 3 - Microservice IA (optionnel mais recommande)

cd smarttask-ai && .venv\Scripts\activate && uvicorn app.main:app --host 0.0.0.0 --port 8000

Entrainement modeles : python -m app.ml.train_all

Terminal 4 - Frontend Angular

cd smarttask-frontend && npm start

Ouvrir navigateur : http://localhost:4200

1.3 Comptes de test

Role	Email	Mot de passe	Dashboard
ADMIN	admin@smarttask.com	Admin123!	/admin/dashboard
MANAGER	manager@smarttask.com	Manager123!	/manager/dashboard
OPERATOR	operator1@smarttask.com	Operator123!	/operator/dashboard
DRIVER	driver1@smarttask.com	Driver123!	/driver/dashboard
DRIVER 2	driver2@smarttask.com	Driver123!	/driver/dashboard
CLIENT	client1@smarttask.com	Client123!	/client/dashboard
CLIENT 2	client2@smarttask.com	Client123!	/client/dashboard

1.4 Matrice de couverture

Module	ADMIN	MGR	OPR	DRV	CLT
Auth / Profil	X	X	X	X	X
Gestion users	X				
Commandes		R	X		R
Taches / Missions		X	R	X	
IA / ML		X	P	R	
Automation		X	X		
Suivi livraison		R	R	X	X
Incidents		X	X	X	X
Avis / Reviews					X
Chatbot Copilot	X	X	X	X	X
Analytics		X		R	

2. SCENARIO GLOBAL E2E (PARCOURS COMPLET)

Ce scenario enchaine tous les roles pour valider le cycle de vie complet d'une commande CRITICAL de la creation a l'avis client. Duree estimee : 45-60 min.

Phase A - Operateur : creer commande urgente

E2E-01 - Creation commande CRITICAL

Role : OPERATOR (operator1@smarttask.com)

Etape 1. Login -> redirect /operator/dashboard
Etape 2. Menu Commandes -> Nouvelle commande (/operator/orders/create)
Etape 3. Selectionner client : Ahmed (client1@smarttask.com)
Etape 4. Type produit : choisir un type catalogue ou Autre
Etape 5. Remplir : quantite 2, montant 200, adresse livraison Tunis
Etape 6. Priorite : CRITICAL
Etape 7. Cliquer 'Analyser reussite livraison' (Delivery Success AI)
Etape 8. Verifier affichage score % et facteurs
Etape 9. Cliquer 'Enregistrer' -> toast succes
- [OK attendu] Commande creee statut PENDING. Si automation active : chaine express declenchee.
- [ECHEC si] Formulaire invalide ou erreur 500

Phase B - Operateur : verification et preparation

E2E-02 - Verifier et preparer commande

Role : OPERATOR

Etape 1. Aller /operator/orders/pending
Etape 2. Ouvrir la commande creee
Etape 3. Cliquer Verifier -> statut VERIFIED
Etape 4. Si pas automation : /operator/tasks -> Preparer commande READY
- [OK attendu] Commande READY, tache PREPARED creee (si workflow manuel)

Phase C - Manager : assignation IA

E2E-03 - Assigner chauffeur via IA

Role : MANAGER (manager@smarttask.com)

Etape 1. Login -> /manager/dashboard : verifier KPIs
Etape 2. Aller /manager/tasks, basculer vue Cartes ou Liste
Etape 3. Ouvrir tache PREPARED non assignee
Etape 4. Cliquer 'Assigner recommande (IA)' ou modal IA
Etape 5. Bouton 'Score reussite livraison' si commande liee
Etape 6. Confirmer assignation -> toast succes
- [OK attendu] Tache ASSIGNED au chauffeur recommande (ex: Mohamed driver1)

Phase D - Chauffeur : execution livraison

E2E-04 - Cycle mission chauffeur

Role : DRIVER (driver1@smarttask.com)

Etape 1. Login -> /driver/dashboard : alertes visibles
Etape 2. Aller /driver/tasks -> voir mission assignee
Etape 3. Ouvrir detail /driver/tasks/{id}
Etape 4. Accepter mission -> statut ACCEPTED
Etape 5. Demarrer livraison -> IN_DELIVERY
Etape 6. Mettre a jour GPS (latitude/longitude)

Etape 7. Soumettre preuve livraison (image)
Etape 8. Terminer livraison -> COMPLETED / DELIVERED
- [OK attendu] Mission terminee, commande DELIVERED

Phase E - Client : suivi et avis

E2E-05 - Suivi et confirmation client

Role : CLIENT (client1@smarttask.com)

Etape 1. Login -> /client/dashboard
Etape 2. Aller /client/orders -> voir commande
Etape 3. Pendant livraison : /client/delivery/{id} suivi temps reel (SSE)
Etape 4. Apres DELIVERED : Confirmer reception
Etape 5. Aller /client/reviews -> noter 5 etoiles + commentaire
- [OK attendu] Commande RECEIVED, avis enregistre, toast/confirmation OK
- [ECHEC si] Double avis sur meme tache -> erreur

3. TESTS AUTHENTIFICATION ET ADMIN

AUTH-01 - Login reussi tous roles

Role : Tous

Etape 1. Pour chaque compte du tableau 1.3 : login email/password
- [OK attendu] Redirection dashboard correspondant au role
- [ECHEC si] Mauvais mot de passe -> message erreur, reste sur /login

AUTH-02 - Acces refuse mauvais role

Role : CLIENT

Etape 1. Login client1
Etape 2. Tenter URL /manager/tasks dans barre adresse
- [OK attendu] Redirect /access-denied ou 403

AUTH-03 - Deconnexion

Role : Tous

Etape 1. Cliquer Deconnexion dans sidebar
Etape 2. Tenter acceder dashboard sans login
- [OK attendu] Redirect /login

AUTH-04 - Profil utilisateur

Role : Tous

Etape 1. Aller /profile
Etape 2. Modifier nom ou telephone
Etape 3. Changer mot de passe
- [OK attendu] Profil mis a jour, re-login avec nouveau MDP si change

Tests ADMIN

ADM-01 - Lister utilisateurs

Role : ADMIN (admin@smarttask.com)

Etape 1. Login admin
Etape 2. /admin/users
Etape 3. Verifier liste 7+ utilisateurs seed
- [OK attendu] Tableau users avec roles ADMIN/MANAGER/OPERATOR/DRIVER/CLIENT

ADM-02 - Creer utilisateur

Role : ADMIN

Etape 1. Cliquer Nouveau
Etape 2. Email test@test.com, role OPERATOR, MDP Test123!
Etape 3. Sauvegarder
- [OK attendu] Utilisateur cree, login possible

ADM-03 - Desactiver utilisateur

Role : ADMIN

Etape 1. Selectionner user test
Etape 2. Desactiver
Etape 3. Tenter login avec ce compte
- [OK attendu] Login refuse compte inactif

ADM-04 - Changer role

Role : ADMIN

Etape 1. Modifier role user test -> CLIENT

Etape 2. Verifier acces /client/dashboard
- [OK attendu] Redirection client apres re-login

4. TESTS MANAGER (TOUTES FONCTIONNALITES)

4.1 Pages a visiter

Route	Composant
/manager/dashboard	ManagerDashboardPageComponent
/manager/tasks	ManagerTasksPageComponent
/manager/planning	ManagerPlanningPageComponent
/manager/map	ManagerMapPageComponent
/manager/drivers	ManagerDriversPageComponent
/manager/delays	ManagerDelaysPageComponent
/manager/incidents	ManagerIncidentsPageComponent
/manager/notifications	ManagerNotificationsPageComponent
/manager/analytics	ManagerAnalyticsPageComponent
/manager/ai	ManagerAiPageComponent
/manager/automation	ManagerAutomationPageComponent
/manager/settings	ManagerSettingsPageComponent

MGR-01 - Dashboard KPIs

Role : MANAGER

Etape 1. Login manager
Etape 2. /manager/dashboard
- [OK attendu] Cartes KPI : taches, retards, chauffeurs. Banniere IA visible.

MGR-02 - Gestion taches CRUD

Role : MANAGER

Etape 1. /manager/tasks
Etape 2. Nouvelle tache
Etape 3. Remplir titre, client, adresse, priorite
Etape 4. Sauvegarder
Etape 5. Modifier tache
Etape 6. Changer statut
Etape 7. Supprimer tache test
- [OK attendu] CRUD complet, filtres statut/recherche OK, vue cartes/liste

MGR-03 - Assignation manuelle

Role : MANAGER

Etape 1. Ouvrir tache PREPARED
Etape 2. Assigner chauffeur driver1 manuellement
- [OK attendu] Tache ASSIGNED, driver1 voit mission

MGR-04 - Assignation IA

Role : MANAGER

Etape 1. Tache PREPARED
Etape 2. Bouton Assigner recommande (IA)
- [OK attendu] Toast succes, chauffeur assigne selon score IA

MGR-05 - Page retards

Role : MANAGER

Etape 1. /manager/delays
- [OK attendu] Liste taches en retard ou message vide

MGR-06 - Chauffeurs

Role : MANAGER

Etape 1. /manager/drivers
- [OK attendu] Liste driver1, driver2 avec statut zone performance

MGR-07 - Incidents

Role : MANAGER

Etape 1. /manager/incidents
Etape 2. Ouvrir incident OPEN
Etape 3. Passer IN_PROGRESS puis RESOLVED
- [OK attendu] Statut incident mis a jour

MGR-08 - Notifications

Role : MANAGER

Etape 1. /manager/notifications
Etape 2. Marquer comme lu
- [OK attendu] readFlag true apres action

MGR-09 - Analytics + export

Role : MANAGER

Etape 1. /manager/analytics
Etape 2. Consulter graphiques
Etape 3. Exporter CSV
- [OK attendu] Fichier CSV telecharge

MGR-10 - Planning et carte

Role : MANAGER

Etape 1. /manager/planning
Etape 2. /manager/map
- [OK attendu] Pages chargent sans erreur console

4.2 Module IA Manager (/manager/ai)

MGR-IA-01 - Sante service IA

Role : MANAGER

Etape 1. /manager/ai
Etape 2. Verifier statut health et modeles
- [OK attendu] Health UP, etat modeles affiche (LOADED ou FALLBACK)

MGR-IA-02 - Prediction retard

Role : MANAGER

Etape 1. Selectionner tache
Etape 2. Analyser retard
- [OK attendu] Probabilite retard + niveau risque LOW/MEDIUM/HIGH

MGR-IA-03 - Prediction ETA

Role : MANAGER

Etape 1. Prediction temps livraison
- [OK attendu] Minutes estimees affichees

MGR-IA-04 - Delivery Success

Role : MANAGER

Etape 1. Modal tache
Etape 2. Score reussite livraison
- [OK attendu] Pourcentage succes + facteurs

MGR-IA-05 - Prioriser tache

Role : MANAGER

Etape 1. Prioriser via IA
- [OK attendu] Score priorite + explication

MGR-IA-06 - Recommander chauffeur

Role : MANAGER

Etape 1. Recommandation chauffeur pour tache
- [OK attendu] Liste candidats scores decroissants

MGR-IA-07 - Detecter anomalie

Role : MANAGER

Etape 1. Detection anomalie sur tache
- [OK attendu] Flag anomalie + details

MGR-IA-08 - Analyse complete

Role : MANAGER

Etape 1. Analyse IA complete tache
- [OK attendu] Resultat combine delay + driver + priority

4.3 Automation (/manager/automation)

MGR-AU-01 - Statut automation

Role : MANAGER

Etape 1. /manager/automation
- [OK attendu] Statut enabled/disabled, config express visible

MGR-AU-02 - Lancer chaine express manuelle

Role : MANAGER

Etape 1. Commande HIGH/CRITICAL
Etape 2. Lancer automation express
Etape 3. Consulter logs
- [OK attendu] 7 etapes dans automation_logs : SUCCESS ou SKIPPED

5. TESTS OPERATEUR

OPR-01 - Dashboard

Role : OPERATOR

Etape 1. /operator/dashboard
- [OK attendu] KPI commandes jour, pending, verified, banniere nouveautes

OPR-02 - Liste commandes vues

Role : OPERATOR

Etape 1. /operator/orders
Etape 2. Basculer Cartes / Liste
- [OK attendu] Commandes affichees, toggle fonctionne

OPR-03 - Creer commande LOW

Role : OPERATOR

Etape 1. /operator/orders/create
Etape 2. Priorite LOW
Etape 3. Enregistrer
- [OK attendu] Commande PENDING, pas d'automation auto

OPR-04 - Creer commande HIGH + IA preview

Role : OPERATOR

Etape 1. Priorite HIGH
Etape 2. Analyser reussite livraison avant save
Etape 3. Enregistrer
- [OK attendu] Score IA affiche, automation peut se declencher

OPR-05 - Commandes en attente

Role : OPERATOR

Etape 1. /operator/orders/pending
- [OK attendu] Liste PENDING uniquement

OPR-06 - Verifier commande

Role : OPERATOR

Etape 1. Ouvrir commande PENDING
Etape 2. Verifier
- [OK attendu] Statut VERIFIED

OPR-07 - Preparer commande

Role : OPERATOR

Etape 1. /operator/tasks
Etape 2. Preparer commande READY
- [OK attendu] Tache PREPARED liee

OPR-08 - Suivi taches

Role : OPERATOR

Etape 1. /operator/tasks/tracking
- [OK attendu] Statuts livraisons visibles

OPR-09 - Signaler incident

Role : OPERATOR

Etape 1. /operator/incidents
Etape 2. Nouveau incident

Etape 3. Type + description

- [OK attendu] Incident OPEN cree

OPR-10 - Liste clients

Role : OPERATOR

Etape 1. Via API ou UI clients disponibles a la creation commande

- [OK attendu] client1, client2 selectionnables

6. TESTS CHAUFFEUR

DRV-01 - Dashboard + alertes

Role : DRIVER

Etape 1. /driver/dashboard
- [OK attendu] Missions du jour, alertes IA si risque retard

DRV-02 - Liste missions

Role : DRIVER

Etape 1. /driver/tasks
- [OK attendu] Missions ASSIGNED/ACCEPTED visibles

DRV-03 - Detail mission workflow

Role : DRIVER

Etape 1. /driver/tasks/{id}
Etape 2. Accept -> Start -> Delivery -> Complete
- [OK attendu] Transitions statut OK, historique mis a jour

DRV-04 - Mise a jour GPS

Role : DRIVER

Etape 1. Mettre a jour position lat/lng
- [OK attendu] Coordonnees sauvegardees sur tache

DRV-05 - Preuve livraison

Role : DRIVER

Etape 1. Upload image preuve
- [OK attendu] proofOfDeliveryImage enregistre

DRV-06 - Itineraire optimise IA

Role : DRIVER

Etape 1. /driver/route
Etape 2. Consulter ordre optimise
- [OK attendu] Missions reordonnees si multi-stops

DRV-07 - Historique

Role : DRIVER

Etape 1. /driver/history
- [OK attendu] Livraisons passees listees

DRV-08 - Performance

Role : DRIVER

Etape 1. /driver/performance
- [OK attendu] Score performance, total livraisons

DRV-09 - Incident chauffeur

Role : DRIVER

Etape 1. /driver/incidents
Etape 2. Signaler CLIENT_ABSENT ou VEHICLE_ISSUE
- [OK attendu] Incident cree, manager notifie

DRV-10 - Alertes proactives

Role : DRIVER

Etape 1. Attendre 5 min ou POST /api/driver/alerts/check

Etape 2. Verifier dashboard
- [OK attendu] Notification si deadline proche + risque retard

7. TESTS CLIENT

CLT-01 - Dashboard

Role : CLIENT

Etape 1. /client/dashboard
- [OK attendu] Resume commandes actives et livrees

CLT-02 - Mes commandes

Role : CLIENT

Etape 1. /client/orders
- [OK attendu] Uniquement commandes du client connecte (client1)

CLT-03 - Detail commande

Role : CLIENT

Etape 1. /client/orders/{id}
- [OK attendu] Statut, adresse, priorite, taskId si lie

CLT-04 - Suivi livraison SSE

Role : CLIENT

Etape 1. /client/delivery/{id} pendant IN_DELIVERY
- [OK attendu] Mises a jour temps reel statut/GPS

CLT-05 - Confirmer reception

Role : CLIENT

Etape 1. Apres DELIVERED
Etape 2. Confirmer reception
- [OK attendu] Statut RECEIVED, clientConfirmed true

CLT-06 - Laisser avis

Role : CLIENT

Etape 1. /client/reviews
Etape 2. Note 4/5 + commentaire
- [OK attendu] Review enregistre

CLT-07 - Incident client

Role : CLIENT

Etape 1. /client/incidents
Etape 2. Signaler DAMAGED_PRODUCT
- [OK attendu] Incident OPEN

CLT-08 - Notifications

Role : CLIENT

Etape 1. /client/notifications
- [OK attendu] Notifications livraison visibles

CLT-09 - Isolation donnees

Role : CLIENT

Etape 1. Login client2
Etape 2. Verifier ne voit pas commandes client1
- [OK attendu] Donnees isolees par clientId

8. TESTS TRANSVERSAUX

8.1 Chatbot Copilot

CHAT-01 - Ouvrir widget

Role : Tous roles

Etape 1. Cliquer bouton flottant Copilot
Etape 2. Verifier init
- [OK attendu] Fenetre chat ouverte, suggestions affichees

CHAT-02 - Poser question metier

Role : MANAGER

Etape 1. Question : Comment assigner une tache ?
Etape 2. Envoyer
- [OK attendu] Reponse contextuelle SmartLogix

CHAT-03 - Historique

Role : Tous

Etape 1. Envoyer 2 messages
Etape 2. Rouvrir widget
- [OK attendu] Historique conserve session

8.2 Toasts et alertes UI

UI-01 - Toast succes

Role : OPERATOR

Etape 1. Creer commande
- [OK attendu] Toast vert 'Commande creee avec succes'

UI-02 - Toast IA

Role : MANAGER

Etape 1. Assigner recommande IA
- [OK attendu] Toast info avec message assignation

UI-03 - Alert banner

Role : OPERATOR

Etape 1. /operator/orders/create
- [OK attendu] Banniere Delivery Success AI visible

8.3 Parametres

SET-01 - Settings par role

Role : Tous

Etape 1. /manager/settings
Etape 2. /operator/settings
Etape 3. /driver/settings
Etape 4. /client/settings
- [OK attendu] Pages parametres chargent

9. TESTS API (POSTMAN / CURL)

Collection Postman : Backend Smarttask/SmartTask/postman/SmartTask-API.postman_collection.json

9.1 Sequence API type

Etape 1. POST /api/auth/login {"email":"manager@smarttask.com","password":"Manager123!"}
Etape 2. Copier token JWT de la reponse
Etape 3. Header Authorization: Bearer {token} sur requetes suivantes
Etape 4. GET /api/manager/dashboard -> 200 + JSON KPIs
Etape 5. GET /api/tasks -> 200 + liste paginee

9.2 Tests securite API

API-SEC-01 - Sans token

Role : N/A

Etape 1. GET /api/manager/dashboard sans Authorization
- [OK attendu] 401 Unauthorized

API-SEC-02 - Mauvais role

Role : CLIENT token

Etape 1. GET /api/tasks avec token client
- [OK attendu] 403 Forbidden

API-SEC-03 - Endpoint public

Role : N/A

Etape 1. POST /api/auth/login
- [OK attendu] 200 avec credentials valides

9.3 Tests service IA direct (backend requis)

Methode	Endpoint	Test
GET	/api/ai/health	Sante service + version
GET	/api/ai/model/status	Etat chargement modeles ML
GET	/api/ai/metrics	Compteurs predictions
POST	/api/ai/predict/delay	Classification risque retard
POST	/api/ai/predict/delivery-time	Regression ETA minutes
POST	/api/ai/detect/anomaly	Detection anomalies statistiques
POST	/api/ai/recommend/driver	Scoring chauffeur rule-based
POST	/api/ai/prioritize/task	Score priorite tache
POST	/api/ai/chat	Reponse chatbot Copilot
GET	/api/ai/chat/suggestions	Suggestions prompts
POST	/api/ai/optimize/route	Optimisation itineraire TSP

AI-01 - Health Python

Role : N/A

Etape 1. GET http://localhost:8000/api/ai/health
- [OK attendu] status UP, version affichee

AI-02 - IA down fallback

Role : MANAGER

Etape 1. Arreter uvicorn Python
Etape 2. Predict delay sur tache

- [OK attendu] Fallback rule-based, pas crash 500 backend

10. TESTS NEGATIFS ET LIMITES

NEG-01 - Login email invalide

Role : N/A

Etape 1. email: notanemail
Etape 2. password: x
- [OK attendu] Validation formulaire ou 401

NEG-02 - Commande champs vides

Role : OPERATOR

Etape 1. Formulaire create sans adresse
- [OK attendu] Bouton submit desactive ou erreur validation

NEG-03 - Complete sans preuve

Role : DRIVER

Etape 1. Terminer sans upload si requis
- [OK attendu] Erreur metier ou warning

NEG-04 - Double avis

Role : CLIENT

Etape 1. Review deux fois meme tache
- [OK attendu] Erreur contrainte unique

NEG-05 - Assigner chauffeur BUSY

Role : MANAGER

Etape 1. Assigner driver deja surcharge
- [OK attendu] Warning ou score IA reduit

11. CHECKLIST RECETTE FINALE

Cocher chaque item avant validation projet.

Module	Critere	OK	Date	Testeur
Auth	Login/logout 7 comptes OK	[ ]		
Auth	RBAC : acces refuse hors role	[ ]		
Admin	CRUD users complet	[ ]		
Operator	Create/verify/prepare commande	[ ]		
Operator	Delivery Success AI preview	[ ]		
Manager	Dashboard + tasks CRUD	[ ]		
Manager	Assign manuel + IA	[ ]		
Manager	8 fonctions page IA	[ ]		
Manager	Automation express + logs	[ ]		
Manager	Analytics export CSV	[ ]		
Driver	Workflow accept->complete	[ ]		
Driver	Route optimisee IA	[ ]		
Driver	Preuve livraison	[ ]		
Client	Suivi SSE	[ ]		
Client	Confirm + review	[ ]		
Transversal	Chatbot 3 roles	[ ]		
Transversal	Toasts et alert banners	[ ]		
API	Postman collection 100% 2xx sur happy path	[ ]		
IA	Health UP ou fallback graceful	[ ]		
Securite	401/403 verifies	[ ]		

11.1 Synthese comptes par scenario

Scenario	Compte principal	Comptes secondaires
E2E complet	operator1	manager, driver1, client1
Automation express	operator1	manager
Multi chauffeur	manager	driver1, driver2
Isolation client	client1 vs client2	
Admin users	admin	user test cree
IA sans Python	manager	service IA arrete

12. DEPANNAGE ET FAQ TESTS

Probleme	Cause probable	Solution
Login 401	Backend off ou mauvais MDP	Verifier bootRun + comptes section 1.3
CORS error	Frontend pas localhost:4200	CORS_ORIGINS dans application.properties
IA 503	Modeles .pkl absents	cd smarttask-ai && python -m app.ml.train_all
IA timeout	Python non demarre	uvicorn port 8000
Empty dashboard	DB vide	Redemarrer backend pour DataInitializer
SSE ne connecte pas	Pas en IN_DELIVERY	Driver start livraison d'abord
403 access denied	Mauvais role URL	Utiliser compte du bon role
Automation skip	Priorite LOW	Utiliser HIGH ou CRITICAL

13. MATRICE TESTS API PAR ENDPOINT

Cocher apres test Postman ou curl. Token JWT requis sauf PUBLIC.

ID	HTTP	Endpoint	Role	OK
API-001	POST	/api/auth/register	PUBLIC	[ ]
API-002	POST	/api/auth/login	PUBLIC	[ ]
API-003	POST	/api/auth/logout	PUBLIC	[ ]
API-004	GET	/api/auth/me	AUTH	[ ]
API-005	GET	/api/users/me	AUTH	[ ]
API-006	PUT	/api/users/me	AUTH	[ ]
API-007	PATCH	/api/users/me/password	AUTH	[ ]
API-008	GET	/api/users	ADMIN	[ ]
API-009	GET	/api/users/{id}	ADMIN	[ ]
API-010	POST	/api/users	ADMIN	[ ]
API-011	PUT	/api/users/{id}	ADMIN	[ ]
API-012	DELETE	/api/users/{id}	ADMIN	[ ]
API-013	PATCH	/api/users/{id}/status	ADMIN	[ ]
API-014	PATCH	/api/users/{id}/role	ADMIN	[ ]
API-015	GET	/api/manager/dashboard	MANAGER	[ ]
API-016	GET	/api/manager/drivers	MANAGER	[ ]
API-017	GET	/api/manager/drivers/{id}	MANAGER	[ ]
API-018	GET	/api/manager/operators	MANAGER	[ ]
API-019	GET	/api/manager/clients	MANAGER	[ ]
API-020	GET	/api/manager/delays	MANAGER	[ ]
API-021	GET	/api/manager/notifications	MANAGER	[ ]
API-022	PATCH	/api/manager/notifications/{id}/read	MANAGER	[ ]

ID	HTTP	Endpoint	Role	OK
API-023	GET	/api/manager/analytics	MANAGER	[ ]
API-024	GET	/api/manager/analytics/export	MANAGER	[ ]
API-025	GET	/api/manager/ai/recommendations	MANAGER	[ ]
API-026	GET	/api/manager/ai/recommend-driver/{taskId}	MANAGER	[ ]
API-027	GET	/api/manager/incidents	MANAGER	[ ]
API-028	PATCH	/api/manager/incidents/{id}	MANAGER	[ ]
API-029	GET	/api/manager/ai/health	MANAGER	[ ]
API-030	GET	/api/manager/ai/model-status	MANAGER	[ ]
API-031	POST	/api/manager/ai/predict-delay	MANAGER	[ ]
API-032	POST	/api/manager/ai/predict-delivery-time	MANAGER	[ ]
API-033	POST	/api/manager/ai/prioritize-task	MANAGER	[ ]
API-034	POST	/api/manager/ai/recommend-driver	MANAGER	[ ]
API-035	POST	/api/manager/ai/detect-anomaly	MANAGER	[ ]
API-036	POST	/api/manager/ai/analyze	MANAGER	[ ]
API-037	GET	/api/tasks	MANAGER	[ ]
API-038	GET	/api/tasks/{id}	MANAGER	[ ]
API-039	GET	/api/tasks/{id}/history	MANAGER	[ ]
API-040	POST	/api/tasks	MANAGER	[ ]
API-041	PUT	/api/tasks/{id}	MANAGER	[ ]

API-042	DELETE	/api/tasks/{id}	MANAGER	[ ]
API-043	PATCH	/api/tasks/{id}/status	MANAGER	[ ]
API-044	PATCH	/api/tasks/{id}/priority	MANAGER	[ ]

ID	HTTP	Endpoint	Role	OK
API-045	POST	/api/tasks/{id}/assign/{driverId}	MANAGER	[ ]
API-046	POST	/api/tasks/{id}/assign-recommended	MANAGER	[ ]
API-047	GET	/api/operator/dashboard	OPERATOR	[ ]
API-048	GET	/api/operator/orders	OPERATOR	[ ]
API-049	GET	/api/operator/orders/pending	OPERATOR	[ ]
API-050	GET	/api/operator/orders/{id}	OPERATOR	[ ]
API-051	POST	/api/operator/orders	OPERATOR	[ ]
API-052	PUT	/api/operator/orders/{id}	OPERATOR	[ ]
API-053	POST	/api/operator/orders/{id}/verify	OPERATOR	[ ]
API-054	GET	/api/operator/tasks	OPERATOR	[ ]
API-055	POST	/api/operator/tasks/{orderId}/prepare	OPERATOR	[ ]
API-056	GET	/api/operator/clients	OPERATOR	[ ]
API-057	GET	/api/operator/incidents	OPERATOR	[ ]
API-058	POST	/api/operator/incidents	OPERATOR	[ ]
API-059	GET	/api/driver/dashboard	DRIVER	[ ]
API-060	GET	/api/driver/tasks	DRIVER	[ ]
API-061	GET	/api/driver/tasks/{id}	DRIVER	[ ]
API-062	PATCH	/api/driver/tasks/{id}/accept	DRIVER	[ ]
API-063	PATCH	/api/driver/tasks/{id}/start	DRIVER	[ ]
API-064	PATCH	/api/driver/tasks/{id}/delivery	DRIVER	[ ]
API-065	PATCH	/api/driver/tasks/{id}/complete	DRIVER	[ ]
API-066	PATCH	/api/driver/tasks/{id}/location	DRIVER	[ ]

ID	HTTP	Endpoint	Role	OK
API-067	POST	/api/driver/tasks/{id}/proof	DRIVER	[ ]
API-068	GET	/api/driver/route	DRIVER	[ ]
API-069	GET	/api/driver/route/optimized	DRIVER	[ ]
API-070	POST	/api/driver/alerts/check	DRIVER	[ ]
API-071	POST	/api/driver/incidents	DRIVER	[ ]
API-072	GET	/api/driver/history	DRIVER	[ ]
API-073	GET	/api/driver/performance	DRIVER	[ ]
API-074	GET	/api/client/dashboard	CLIENT	[ ]
API-075	GET	/api/client/orders	CLIENT	[ ]
API-076	GET	/api/client/orders/{id}	CLIENT	[ ]
API-077	GET	/api/client/deliveries/{id}	CLIENT	[ ]
API-078	GET	/api/client/deliveries/{id}/stream	CLIENT	[ ]
API-079	POST	/api/client/deliveries/{id}/confirm	CLIENT	[ ]
API-080	POST	/api/client/incidents	CLIENT	[ ]
API-081	POST	/api/client/reviews	CLIENT	[ ]
API-082	GET	/api/client/reviews	CLIENT	[ ]
API-083	GET	/api/client/notifications	CLIENT	[ ]
API-084	GET	/api/automation/status	MGR/OP	[ ]
API-085	POST	/api/automation/express/{orderId}	MGR/OP	[ ]
API-086	GET	/api/automation/logs	MANAGER	[ ]

API-087	GET	/api/automation/logs/run/{runId}	MGR/OP	[ ]
API-088	GET	/api/product-types	AUTH	[ ]

ID	HTTP	Endpoint	Role	OK
API-089	GET	/api/chat/init	AUTH	[ ]
API-090	POST	/api/chat/message	AUTH	[ ]
API-091	GET	/api/chat/history	AUTH	[ ]

14. MATRICE TESTS UI PAR ROUTE ANGULAR

Route	Composant	Role	OK
/login	LoginComponent	Public	[ ]
/access-denied	AccessDeniedComponent	Public	[ ]
/admin/dashboard	AdminDashboardComponent	ADMIN	[ ]
/admin/users	AdminUsersComponent	ADMIN	[ ]
/manager/dashboard	ManagerDashboardPageComponent	MANAGER	[ ]
/manager/tasks	ManagerTasksPageComponent	MANAGER	[ ]
/manager/planning	ManagerPlanningPageComponent	MANAGER	[ ]
/manager/map	ManagerMapPageComponent	MANAGER	[ ]
/manager/drivers	ManagerDriversPageComponent	MANAGER	[ ]
/manager/delays	ManagerDelaysPageComponent	MANAGER	[ ]
/manager/incidents	ManagerIncidentsPageComponent	MANAGER	[ ]
/manager/notifications	ManagerNotificationsPageComponent	MANAGER	[ ]
/manager/analytics	ManagerAnalyticsPageComponent	MANAGER	[ ]
/manager/ai	ManagerAiPageComponent	MANAGER	[ ]
/manager/automation	ManagerAutomationPageComponent	MANAGER	[ ]
/manager/settings	ManagerSettingsPageComponent	MANAGER	[ ]
/operator/dashboard	OperatorDashboardPageComponent	OPERATOR	[ ]
/operator/orders	OperatorOrdersPageComponent	OPERATOR	[ ]
/operator/orders/create	OperatorOrderCreatePageComponent	OPERATOR	[ ]
/operator/orders/pending	OperatorPendingPageComponent	OPERATOR	[ ]

Route	Composant	Role	OK
/operator/tasks	OperatorTasksPageComponent	OPERATOR	[ ]
/operator/tasks/tracking	OperatorTrackingPageComponent	OPERATOR	[ ]
/operator/incidents	OperatorIncidentsPageComponent	OPERATOR	[ ]
/operator/settings	OperatorSettingsPageComponent	OPERATOR	[ ]
/driver/dashboard	DriverDashboardPageComponent	DRIVER	[ ]
/driver/tasks	DriverTasksPageComponent	DRIVER	[ ]
/driver/tasks/:id	DriverTaskDetailPageComponent	DRIVER	[ ]
/driver/route	DriverRoutePageComponent	DRIVER	[ ]
/driver/incidents	DriverIncidentsPageComponent	DRIVER	[ ]
/driver/history	DriverHistoryPageComponent	DRIVER	[ ]
/driver/performance	DriverPerformancePageComponent	DRIVER	[ ]
/driver/settings	DriverSettingsPageComponent	DRIVER	[ ]
/client/dashboard	ClientDashboardPageComponent	CLIENT	[ ]
/client/orders	ClientOrdersPageComponent	CLIENT	[ ]
/client/orders/:id	ClientOrderDetailPageComponent	CLIENT	[ ]
/client/delivery/:id	ClientDeliveryPageComponent	CLIENT	[ ]
/client/notifications	ClientNotificationsPageComponent	CLIENT	[ ]
/client/reviews	ClientReviewsPageComponent	CLIENT	[ ]
/client/incidents	ClientIncidentsPageComponent	CLIENT	[ ]
/client/settings	ClientSettingsPageComponent	CLIENT	[ ]

Route	Composant	Role	OK
/profile	ProfileComponent	AUTH	[ ]